

JPHG01 Temperature and thermometer — Quiz 3

S3 Physics · 3 multiple-choice items

Q1 · JPHG01.2 · Standard

The resistance of the metal coil in a resistance thermometer is $70\ \Omega$ at $30\ ^\circ\text{C}$ and $80\ \Omega$ at $50\ ^\circ\text{C}$. What is the resistance of the metal coil if it is placed in boiling water? Assume that the resistance of the coil varies linearly with temperature.

- A. $105\ \Omega$
- B. $150\ \Omega$
- C. $155\ \Omega$
- D. $205\ \Omega$

Q2 · JPHG01.3 · Foundation

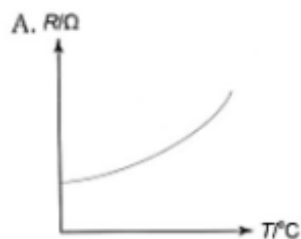
A mercury-in-glass thermometer will response to temperature change more quickly if

- A. the inner diameter of the tube is reduced.
- B. the thickness of the wall is reduced.
- C. the size of the bulb is increased.
- D. the length of the tube is increased.

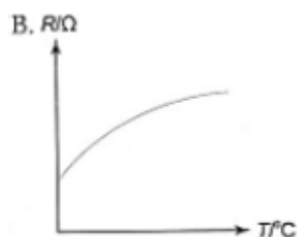
Q3 · JPHG01.3 · Standard

The graphs below show the variation of the electrical resistance R of four circuit elements with temperature T . Which of the circuit elements is the most suitable for measuring temperature?

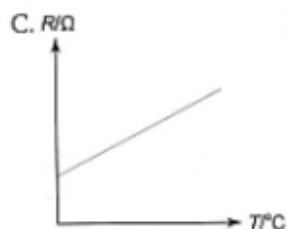
A. R increases with T , and the graph curves upwards.



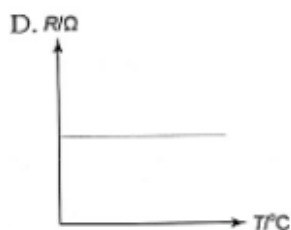
B. R increases with T , and the graph curves downwards.



C. R increases linearly with T .



D. R does not change with T .



JPHG01 Temperature and thermometer — Quiz 3 · Answer key

Teacher page · 3 items

Q1 · JPHG01.2

Answer: A. 105 Ω

Guide: Let the resistance of the metal coil at 100 °C be x.

$$(x - 70)/(80 - 70) = (100 - 30)/(50 - 30)$$

$$x = 105 \Omega.$$

Q2 · JPHG01.3

Answer: B. the thickness of the wall is reduced.

Q3 · JPHG01.3

Answer: C. R increases linearly with T.

Guide: The electrical resistance R of the circuit element varies linearly with the temperature T. The thermometer making use of the variation of R can have a linear scale which is the most suitable.